E-Bayesian Estimation for the Lomax Distribution Under the Precautionary Loss Function Based on Record Values

S.M.T.K. MirMostafaee[1]

Department of Statistics, University of Mazandaran, P.O. Box 47416-1467, Babolsar, Iran

---

Abstract:

This paper addresses E-Bayesian estimation of the Lomax distribution parameter within the framework of the precautionary loss function, using upper record values. The E-posterior risks of the proposed estimators are derived. A simulation study is carried out to compare and evaluate the estimators. A simulated data example is also given. The paper concludes with a discussion.

Keywords: Precautionary loss function, E-Bayesian estimation, Upper record values, and E-posterior risk.

Mathematics Subject Classification (2010): 62F10.

---

1 Introduction

The Lomax distribution has been proved to be useful in engineering, industry and medical science (Okasha et al., 2021). Suppose X follows a Lomax distribution with shape parameter θ. We denote this by $X \sim \text{Lomax}(\theta)$. The probability density function (PDF), and cumulative distribution function(CDF) of X are $f(x;\theta) = \theta\,(1+x)^{-(\theta+1)}$, and $F(x;\theta) = 1-(1+x)^{-\theta}$, respectively, where $x > 0$, and $\theta > 0$.

Record values arise naturally in many practical settings. Consider a sequence $\{X_i; i \geq 1\}$ of

[1]Corresponding Author: m.mirmostafaee@umz.ac.ir

independent and identically distributed continuous random variables with common PDF $f(x)$ and CDF $F(x)$. An observation X_j is said to be an upper record value if it is greater than all previous observations $X_1, \dots, X_{j-1}$. By convention, X_1 is taken as the first lower record. If R_n denotes the nth upper record value, then the joint PDF of the first n upper records, $R_1, \cdots R_m$, is also given by (see Arnold et al. (1998))

$$f_{R_1,\cdots,R_m}(r_1, \cdots, r_m) = f(r_m) \prod_{i=1}^{m-1} \frac{f(r_i)}{1 - F(r_i)}, \quad r_1 < \cdots < r_m. \tag{1.1}$$

Lower record values are defined similarly. For further details on the theory and applications of record values, see Arnold et al. (1998).

Bayesian inference generally requires specifying a prior distribution and a loss function. Here, we adopt the precautionary loss function (PLF), given by Norstorm (1996), defined as $L(\theta, \delta) = \dfrac{(\theta - \delta)^2}{\delta}$, where δ denotes an estimator of θ. Although some practitioners assume that a single prior distribution can be elicited, prior information is frequently vague in practice, meaning that any elicited prior is merely an approximation (see Alhamaidah et al. (2023)). A useful approach to address this issue is the E-Bayesian method introduced by Han (1997); see also Alhamaidah et al. (2023) for a recent study.

In the following sections, we derive the E-Bayesian estimator for the Lomax parameter under the precautionary loss function using upper record data. The associated E-posterior risks are also obtained. A simulation study is conducted to compare and assess the performance of the estimators. An example based on simulated data is also presented.

2 Main Results

Let $\boldsymbol{R} = (R_1, \dots, R_m)$ represent the vector of the first m upper record values from a Lomax distribution with parameter θ. Then, from (1.1), the joint PDF of $R_1, \dots, R_m$ is

$$f_{R_1,\cdots,R_m}(r_1, \cdots, r_m; \theta) = \theta^m \exp\big(-\theta \ln(1 + r_m)\big) \prod_{i=1}^{m} \frac{1}{1 + r_i}, \quad r_1 < \cdots < r_m. \tag{2.1}$$

The maximum likelihood (ML) estimate of θ is given by $\hat{\theta}_M = \dfrac{m}{\ln(1 + r_m)}$, found by maximizing (2.1) with respect to θ.

For Bayesian inference, we assume the following gamma prior for the parameter θ

$$\pi(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} \mathrm{e}^{-b\theta}, \qquad \theta > 0, \quad a > 0, b > 0. \tag{2.2}$$

Using (2.1) and (2.2), the posterior density of θ given the observed records $\boldsymbol{r} = (r_1, \cdots, r_m)$ is obtained as a gamma distribution with parameters $m+a$ and $b+\ln(1+r_m)$, which is given by

$$\pi(\theta|\boldsymbol{r}) = \frac{[b+\ln(1+r_m)]^{m+a}}{\Gamma(m+a)}\,\theta^{m+a-1}\mathrm{e}^{-\theta\,[b+\ln(1+r_m)]}.$$

Now, under the PLF, the Bayesian estimator of θ is derived as follows:

$$\widetilde{\theta}_B = \sqrt{E(\theta^2|\boldsymbol{R})} = \frac{\sqrt{(m+a)(m+a+1)}}{b+\ln(1+R_m)}.$$

Next, we obtain the posterior risk of $\widetilde{\theta}_B$, denoted as $PR(\widetilde{\theta}_B)$. We have

$$\begin{aligned} PR(\widetilde{\theta}_B) &= \int_0^\infty \frac{(\theta-\widetilde{\theta}_B)^2}{\widetilde{\theta}_B}\pi(\theta|\boldsymbol{r})\mathrm{d}\theta = \frac{1}{\widetilde{\theta}_B}\int_0^\infty \left(\widetilde{\theta}_B^2+\theta^2-2\,\widetilde{\theta}_B\,\theta\right)\pi(\theta|\boldsymbol{r})\mathrm{d}\theta \\ &= \widetilde{\theta}_B + \frac{E(\theta^2|\boldsymbol{r})}{\widetilde{\theta}_B} - 2\,E(\theta|\boldsymbol{r}) = \frac{2\left(\sqrt{(m+a)(m+a+1)}-(m+a)\right)}{b+\ln(1+r_m)}. \end{aligned} \tag{2.3}$$

2.1 E-Bayesian Estimation

Let $\widetilde{\theta}_B$ denote the Bayes estimator of θ obtained under the prior specified in (2.2), and let $\pi^*(a,b)$ denote the joint prior density of the hyperparameters a and b. Then, as defined by Han (1997), the E-Bayesian estimator of θ is

$$\widetilde{\theta}_{EB} = \iint_{\mathbb{D}} \widetilde{\theta}_B\;\pi^*(a,b)\,\mathrm{d}a\mathrm{d}b,$$

where $\mathbb{D}$ is the domain of a and b.

As suggested by Han (1997), the hyperparameters a and b should be specified to ensure that the prior density is a decreasing function of θ. Differentiating (2.2) shows that this holds for $b>0$ and $0<a<1$. On the other hand, it must be pointed out that excessively large values of b can undermine robustness, leading us to restrict $b \le c$ (Berger, 1985). We thus adopt the following three joint priors for a and b:

$$\pi_1(a,b) = \frac{1}{c}, \quad 0<a<1, \quad 0<b<c, \tag{2.4}$$

$$\pi_2(a,b) = \frac{2(c-b)}{c^2}, \quad 0<a<1, \quad 0<b<c, \tag{2.5}$$

$$\pi_3(a,b) = \frac{2b}{c^2}, \quad 0<a<1, \quad 0<b<c. \tag{2.6}$$

Now, using the prior in (2.4), the E-Bayesian estimator of θ under the PLF is obtained as

$$\begin{aligned} \widetilde{\theta}_{EB_1} &= \int_0^c\int_0^1 \frac{\sqrt{(m+a)(m+a+1)}}{b+\ln(1+R_m)}\times\frac{1}{c}\,\mathrm{d}a\,\mathrm{d}b \\ &= \frac{1}{c}\int_0^c \frac{1}{b+\ln(1+R_m)}\mathrm{d}b\int_0^1\sqrt{(m+a)(m+a+1)}\;\mathrm{d}a \\ &= \frac{1}{c}\ln\left(\frac{c+\ln(1+R_m)}{\ln(1+R_m)}\right)\int_0^1\sqrt{(m+a)(m+a+1)}\;\mathrm{d}a. \end{aligned} \tag{2.7}$$

Similarly, using the priors in (2.5), and (2.6), the E-Bayesian estimators of θ under the PLF are obtained as

$$\begin{aligned}\widetilde{\theta}_{EB_2} &= \frac{2}{c^2}\int_0^c \frac{c-b}{b+\ln(1+R_m)}\,\mathrm{d}b\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a\\ &= \frac{2}{c^2}\left[\left(c+\ln(1+R_m)\right)\ln\left(\frac{c+\ln(1+R_m)}{\ln(1+R_m)}\right)-c\right]\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a,\end{aligned}\tag{2.8}$$

and

$$\begin{aligned}\widetilde{\theta}_{EB_3} &= \frac{2}{c^2}\int_0^c \frac{b}{b+\ln(1+R_m)}\,\mathrm{d}b\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a\\ &= \frac{2}{c^2}\left[c-\ln(1+R_m)\ln\left(\frac{c+\ln(1+R_m)}{\ln(1+R_m)}\right)\right]\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a,\end{aligned}\tag{2.9}$$

respectively.

The E-posterior risk was originally defined by Han (2019) under the squared error loss function, where it was called the expected mean square error (E-MSE). The current terminology was later adopted by Han (2020a) and Han (2020b).

Let $\widetilde{\theta}_B$ be the Bayesian estimator of θ under the PLF, and denote its posterior risk by $PR(\widetilde{\theta}_B)$. Let $\widetilde{\theta}_{EB}$ be the corresponding E-Bayesian estimator, obtained using the joint prior $\pi^*(a,b)$ for the hyperparameters a and b. The E-posterior risk of $\widetilde{\theta}_{EB}$ is then defined as

$$EP(\widetilde{\theta}_{EB}) = \iint_{\mathbb{D}} PR(\widetilde{\theta}_B)\,\pi^*(a,b)\,\mathrm{d}a\,\mathrm{d}b,\tag{2.10}$$

where $\mathbb{D}$ represents the parameter space of (a,b).

Now, from (2.3), and (2.10), the E-posterior risks of the E-Bayesian estimators for parameter θ, given in (2.7), (2.8), and (2.9) under the PLF are given by

$$\begin{aligned}EP(\widetilde{\theta}_{EB_1}) &= \int_0^c\int_0^1 \frac{2\left(\sqrt{(m+a)(m+a+1)}-(m+a)\right)}{b+\ln(1+R_m)}\times\frac{1}{c}\,\mathrm{d}a\,\mathrm{d}b\\ &= \frac{2}{c}\ln\left(\frac{c+\ln(1+R_m)}{\ln(1+R_m)}\right)\left[\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a-\frac{2m+1}{2}\right],\\ EP(\widetilde{\theta}_{EB_2}) &= \frac{4}{c^2}\left[\left(c+\ln(1+R_m)\right)\ln\left(\frac{c+\ln(1+R_m)}{\ln(1+R_m)}\right)-c\right]\left[\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a-\frac{2m+1}{2}\right],\end{aligned}$$

and

$$EP(\widetilde{\theta}_{EB_3}) = \frac{4}{c^2}\left[c-\ln(1+R_m)\ln\left(\frac{c+\ln(1+R_m)}{\ln(1+R_m)}\right)\right]\left[\int_0^1 \sqrt{(m+a)(m+a+1)}\,\mathrm{d}a-\frac{2m+1}{2}\right],$$

respectively.

3 A Simulation Study

We conduct a simulation study with $S = 3000$ replications to evaluate the suggested estimators. We set $\theta \in \{2,3,5\}$, $m \in \{3,4,5\}$, and $c \in \{1,2,5\}$. For Bayesian estimation, following Ghasabani et al. (2026), we set $a = 0.5$, and choose $b = a/\theta$ so that the prior mean equals the true θ. In each replication, we generate m upper records from Lomax(θ) and compute the ML, Bayesian, and E-Bayesian estimators of θ. The estimated risk (ER) of an estimator $\hat{\theta}$ is defined as

$$ER(\hat{\theta}) = \frac{1}{S}\sum_{i=1}^{S}\frac{(\hat{\theta}_i - \theta)^2}{\hat{\theta}_i},$$

where $\hat{\theta}_i$ denotes the estimator obtained in the i-th replication. We also compute the average E-posterior risks (AEPRs) for the E-Bayesian estimators. The associated results are reported in Table 1. From Table 1, we draw the following conclusions:

- The E-Bayesian estimators under prior (2.6) yield the smallest AEPRs, while those under prior (2.5) produce the largest AEPRs.
- In terms of ER, the E-Bayesian estimators for $c = 1$ and $c = 2$ outperform both the ML and Bayesian estimators in most cases. However, for $c = 5$, the E-Bayesian estimators under prior (2.6) yield the largest ERs in all cases, and those under prior (2.4) produce the second largest ERs for most cases.
- The Bayesian estimators consistently outperform the ML estimators in terms of ER across all cases.
- Both ER and AEPR decrease as m increases, as expected. Moreover, AEPR decreases with respect to c.

4 An example

Here, we generate 5 upper records from Lomax(3). The simulated records are 0.4092635, 2.0785400, 3.0170122, 6.4595842, 8.2539605. Under the settings stated in the previous section, we compute the ML, Bayesian, and E-Bayesian estimators of θ. The results are given as follows:

			E-Bayes	$c=1$	
ML	Bayes		$\widetilde{\theta}_{EB_1}$	$\widetilde{\theta}_{EB_2}$	$\widetilde{\theta}_{EB_3}$
2.24714	2.49993		2.21925	2.35622	2.08228
E-Bayes	$c=2$		E-Bayes	$c=5$	
$\widetilde{\theta}_{EB_1}$	$\widetilde{\theta}_{EB_2}$	$\widetilde{\theta}_{EB_3}$	$\widetilde{\theta}_{EB_1}$	$\widetilde{\theta}_{EB_2}$	$\widetilde{\theta}_{EB_3}$
1.91705	2.12054	1.71355	1.40840	1.67868	1.13813

We see that the ML, Bayes and E-Bayesian estimates for $c = 1$ are rather close to the true value of θ in this example.

Table 1: The ERs and AEPRs of the ML, Bayesian and E-Bayesian estimators for different values of θ, c and m.

					E-Bayes $c=1$			E-Bayes $c=2$			E-Bayes $c=5$		
θ	m		ML	Bayes	$\widetilde{\theta}_{EB_1}$	$\widetilde{\theta}_{EB_2}$	$\widetilde{\theta}_{EB_3}$	$\widetilde{\theta}_{EB_1}$	$\widetilde{\theta}_{EB_2}$	$\widetilde{\theta}_{EB_3}$	$\widetilde{\theta}_{EB_1}$	$\widetilde{\theta}_{EB_2}$	$\widetilde{\theta}_{EB_3}$
2	3	ER	1.04021	0.66826	0.47196	0.65044	0.32889	0.32692	0.41708	0.34772	0.66794	0.39372	1.39212
		AEPR			0.59545	0.67328	0.51761	0.46843	0.55693	0.37993	0.30725	0.39370	0.22080
	4	ER	0.67732	0.51802	0.38027	0.48933	0.29277	0.28013	0.33974	0.29113	0.53834	0.33039	1.04901
		AEPR			0.45935	0.50377	0.41492	0.37735	0.43356	0.32115	0.26097	0.32304	0.19889
	5	ER	0.49424	0.41771	0.32131	0.39160	0.26542	0.25163	0.29153	0.26044	0.45810	0.29258	0.84167
		AEPR			0.37157	0.39958	0.34355	0.31497	0.35339	0.27656	0.22708	0.27387	0.18029
3	3	ER	1.56031	1.00239	0.54218	0.75434	0.43454	0.56103	0.51936	0.91672	1.75089	0.89394	3.81503
		AEPR			0.78269	0.91048	0.65489	0.59150	0.72646	0.45654	0.36903	0.48904	0.24903
	4	ER	1.01599	0.77704	0.45778	0.59439	0.38645	0.47055	0.43796	0.70814	1.40137	0.73404	2.89857
		AEPR			0.61930	0.69702	0.54158	0.48830	0.57909	0.39752	0.31981	0.40998	0.22964
	5	ER	0.74136	0.62656	0.40392	0.49504	0.35721	0.41759	0.38853	0.59084	1.17460	0.63428	2.32333
		AEPR			0.51009	0.56148	0.45870	0.41533	0.48025	0.35041	0.28294	0.35351	0.21238
5	3	ER	2.60052	1.67066	0.84107	0.92260	1.15415	1.66986	0.98429	3.48030	5.68379	2.88319	12.54792
		AEPR			1.06830	1.29277	0.84383	0.76812	0.98425	0.55200	0.45238	0.62446	0.28029
	4	ER	1.69331	1.29506	0.71593	0.76837	0.91820	1.34585	0.82597	2.62253	4.58687	2.34555	9.68886
		AEPR			0.87239	1.01963	0.72516	0.65242	0.80761	0.49723	0.40071	0.53650	0.26492
	5	ER	1.23560	1.04426	0.64231	0.67319	0.78900	1.14524	0.73145	2.10417	3.84716	1.99152	7.84223
		AEPR			0.73579	0.83898	0.63259	0.56771	0.68468	0.45074	0.36113	0.47187	0.25039

Discussion and Conclusion

In this paper, we studied E-Bayesian estimation of the Lomax distribution parameter under the precautionary loss function, based on upper record values. We obtained the corresponding E-posterior risks for the E-Bayesian estimators. The simulation study demonstrated that the E-Bayesian estimators can perform better than the ML and Bayesian estimators with respect to estimated risk. All numerical results were computed using R (R Core Team, 2026).

Acknowledgments

AI tools were used only for language editing. The author is responsible for the scientific content and conclusions.

References

Alhamaidah, A., Naghizadeh Qomi, M. and Kiapour, A. (2023), Comparison of E-Bayesian estimators in Burr XII model using E-PMSE based on record values, Statistics, Optimization & Information Computing, 11(3), 709-718.

Arnold, B.C., Balakrishnan, N. and Nagaraja, H.N. (1998), Records, New York, John Wiley & Sons.

Berger, J.O. (1985), Statistical decision theory and Bayesian analysis, 2nd edn., New York, Springer-Verlag.

Ghasabani, R., MirMostafaee, S.M.T.K., Naghizadeh Qomi, M. and Basiri, E. (2026), E-Bayesian estimation in the proportional hazard rate family based on progressively Type II censored data with fixed and random sample sizes, Journal of the Indian Society for Probability and Statistics, 27(1), 61-107.

Han, M. (1997), The structure of hierarchical prior distribution and its applications, Chinese Operations Research and Management Science, 6(3), 31-40.

Han, M. (2019), E-Bayesian estimation and its E-MSE under the scaled squared error loss function, for exponential distribution as example. Communications in Statistics-Simulation and Computation, 48(6), 1880-1890.

Han, M. (2020a), E-Bayesian estimation and its E-posterior risk of the exponential distribution parameter based on complete and type I censored samples, Communications in Statistics-Theory and Methods, 49(8), 1858-1872.

Han, M. (2020b), E-Bayesian estimations of the reliability and its E-posterior risk under different loss functions, Communications in Statistics-Simulation and Computation, 49(6), 1527-1545.

Norstorm, J.G. (1996), The use of precautionary loss function in risk analysis, IEEE Transactions on Reliability, 45(3), 400-403.

Okasha, H., Lio, Y. and Albassam, M. (2021), On reliability estimation of Lomax distribution under adaptive type-I progressive hybrid censoring scheme, Mathematics, 9(22), 2903.

R Core Team (2026), R: A language and environment for statistical computing, R Foundation for Statistical Computing, Vienna, Austria.